HIGH-LEVEL DESIGN

Battery Management System High-Level Design

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Classification: Synthetic training data - no real vehicle program

1 Introduction

This document describes the high-level software design of the battery management functions of an electric drive system: cell monitoring, state of charge estimation, thermal management and charge control.

The design is synthetic and was created only for training and evaluation of an AI document analysis assistant.

2 System Overview

The battery management software protects the cells against over-voltage, over-temperature and over-current and supplies the charger with the allowed charge current.

3 Architecture Overview

The software follows the AUTOSAR Classic layered architecture. Safety relevant components are ASIL C or ASIL B and run in a separate OS application with memory protection.

4 Software Component Design

This section describes every application software component with its ports, dependencies and verification cases.

4.1 SWC_CellMonitor

Component: SWC_CellMonitor

Type: Application SWC

Safety Level: ASIL C

Description: Measures the voltage and temperature of every battery cell and publishes the values for the state estimation and thermal management functions.

Design Rationale: Cell level monitoring is safety relevant because overvoltage can lead to a thermal event.

Runnables: Run_CellMonitor_10ms (10 ms)

Provided Ports:

```
- P_CellVoltage | SR_CellVoltage | Sender-Receiver | period 10 ms
- P_CellTemp | SR_CellTemp | Sender-Receiver | period 100 ms
```

Required Ports:

```
- none
```

Depends On: none

Error Handling: A cell measurement outside the plausible range is flagged invalid and the pack is limited.

Verification: TC_BMS_001, TC_BMS_002

4.2 SWC_CurrentSensorAdapter

Component: SWC_CurrentSensorAdapter

Type: Application SWC

Safety Level: ASIL B

Description: Reads the pack current sensor and publishes the filtered pack current.

Design Rationale: The adapter hides the sensor type from the estimation functions.

Runnables: Run_CurrentSensorAdapter_10ms (10 ms)

Provided Ports:

```
- P_PackCurrent | SR_PackCurrent | Sender-Receiver | period 10 ms
```

Required Ports:

```
- none
```
Depends On: none

Error Handling: A sensor offset error is detected during standstill and reported as a diagnostic event.

Verification: TC_BMS_005

4.3 SWC_SocEstimator

Component: SWC_SocEstimator

Type: Application SWC

Safety Level: ASIL B

Description: Estimates the state of charge of the battery pack by combining current integration with a voltage based correction.

Design Rationale: The correction limits the drift of the coulomb counting method.

Runnables: Run_SocEstimator_100ms (100 ms)

Provided Ports:

```
- P_Soc | SR_Soc | Sender-Receiver | period 100 ms
```
Required Ports:

```
- R_CellVoltage | SR_CellVoltage | Sender-Receiver | timeout 30 ms
- R_PackCurrent | SR_PackCurrent | Sender-Receiver | timeout 30 ms
```
Depends On: SWC_CellMonitor

Error Handling: If an input signal is invalid the last valid estimate is held for 2 seconds and then flagged invalid.

Verification: TC_BMS_010

4.4 SWC_ThermalMgr

Component: SWC_ThermalMgr

Type: Application SWC

Safety Level: ASIL B

Description: Controls the cooling and heating of the battery pack based on the cell temperatures and publishes the coolant temperature and the cooling request.

Design Rationale: Early thermal management extends the lifetime of the cells.

Runnables: Run_ThermalMgr_200ms (200 ms)

Provided Ports:

```
- P_CoolantTemp | SR_CoolantTemp | Sender-Receiver | period 200 ms
- P_CoolantReq | SR_CoolantReq | Sender-Receiver | period 200 ms
```
Required Ports:

```
- R_CellTemp | SR_CellTemp | Sender-Receiver | timeout 80 ms
```
Depends On: SWC_CellMonitor

Error Handling: On loss of the cell temperature the cooling is set to maximum.

Verification: TC_BMS_020, TC_BMS_021

4.5 SWC_ChargeCtrl

Component: SWC_ChargeCtrl

Type: Application SWC

Safety Level: ASIL C

Description: Calculates the allowed charge current from the state of charge and the coolant temperature and publishes the charge limit to the charger.

Design Rationale: The charge limit protects the cells against lithium plating at low temperature.

Runnables: Run_ChargeCtrl_50ms (50 ms)

Provided Ports:

```
- P_ChargeLimit | SR_ChargeLimit | Sender-Receiver | period 50 ms
```

Required Ports:

```
- R_Soc | SR_Soc | Sender-Receiver | timeout 150 ms
- R_CoolantTemp | SR_CoolantTemperature | Sender-Receiver | timeout 400 ms
```

Depends On: SWC_SocEstimator, SWC_BmsGateway

Error Handling: If the state of charge is invalid the charge limit is reduced to zero.

Verification: none defined

5 Interface Definitions

All interfaces are Sender-Receiver interfaces unless stated otherwise.

5.1 SR_CellVoltage

Interface: SR_CellVoltage

Paradigm: Sender-Receiver

Data Elements: SIG_CellVoltage

Description: Carries the cell voltages.

5.2 SR_CellTemp

Interface: SR_CellTemp

Paradigm: Sender-Receiver

Data Elements: SIG_CellTemp

Description: Carries the cell temperatures.

5.3 SR_PackCurrent

Interface: SR_PackCurrent

Paradigm: Sender-Receiver

Data Elements: SIG_PackCurrent

Description: Carries the filtered pack current.

5.4 SR_Soc

Interface: SR_Soc

Paradigm: Sender-Receiver

Data Elements: SIG_Soc

Description: Carries the estimated state of charge.

5.5 SR_CoolantTemp

Interface: SR_CoolantTemp

Paradigm: Sender-Receiver

Data Elements: SIG_CoolantTemp

Description: Carries the coolant temperature.

5.6 SR_CoolantReq

Interface: SR_CoolantReq

Paradigm: Sender-Receiver

Data Elements: SIG_CoolantReq

Description: Carries the cooling request.

5.7 SR_ChargeLimit

Interface: SR_ChargeLimit

Paradigm: Sender-Receiver

Data Elements: SIG_ChargeLimit

Description: Carries the allowed charge current.

6 Signal Catalogue

```
Signal | Data Type | Unit | Range | Update Period
SIG_CellVoltage | uint16 | mV | 2500..4300 | 10 ms
SIG_CellTemp | sint16 | 0.1 degC | -400..1250 | 100 ms
SIG_PackCurrent | sint32 | mA | -400000..400000 | 10 ms
SIG_Soc | uint16 | 0.01 percent | 0..10000 | 100 ms
SIG_CoolantTemp | sint16 | 0.1 degC | -400..1250 | 200 ms
SIG_CoolantReq | uint8 | - | 0..3 | 200 ms
SIG_ChargeLimit | uint16 | A | 0..400 | 50 ms
```

7 Functional Flows

The following flows describe how the components cooperate.

7.1 Charge Limit Calculation Flow

Step 1 - SWC_CellMonitor and SWC_CurrentSensorAdapter publish cell voltages and pack current.

Step 2 - SWC_SocEstimator calculates the state of charge every 100 ms.

Step 3 - SWC_ThermalMgr publishes the coolant temperature based on the cell temperatures.

Step 4 - SWC_ChargeCtrl combines the state of charge and the coolant temperature into a charge limit.

7.2 Cooling Request Flow

Step 1 - SWC_ThermalMgr evaluates the highest cell temperature.

Step 2 - When the temperature exceeds 40 degC a cooling request is published.

Step 3 - The request is forwarded to the thermal system through the RTE.

8 Timing and Resource Constraints

Runnables run in 10 ms, 50 ms, 100 ms and 200 ms tasks. The 10 ms task monitors the cell voltage and has the highest priority.

9 Verification Overview

SWC_CellMonitor is verified by TC_BMS_001, TC_BMS_002.

SWC_CurrentSensorAdapter is verified by TC_BMS_005.

SWC_SocEstimator is verified by TC_BMS_010.

SWC_ThermalMgr is verified by TC_BMS_020, TC_BMS_021.

SWC_ChargeCtrl has no verification case defined yet.

10 Revision History

Revision 1.0 - Initial release of the battery management high-level design.